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Inventory Alert Decision Rules

Purpose

This document defines the decision logic for the watsonx Orchestrate demo agent when processing velocity spike alerts. These rules help the agent determine urgency levels and recommend appropriate actions.

Rule Categories

1. Urgency Assessment Rules

Rule 1.1: Velocity Ratio Threshold

```
IF velocity_ratio >= 10.0 THEN urgency_score += 3
ELSE IF velocity_ratio >= 5.0 THEN urgency_score += 2
ELSE IF velocity_ratio >= 3.0 THEN urgency_score += 1
```

Rationale: Higher velocity ratios indicate more significant deviations from normal behavior.

Rule 1.2: Stockout Timeline

```
IF hours_to_stockout <= 4 THEN urgency_score += 3
ELSE IF hours_to_stockout <= 12 THEN urgency_score += 2
ELSE IF hours_to_stockout <= 24 THEN urgency_score += 1
```

Rationale: Imminent stockouts require immediate action.

Rule 1.3: Product Value Impact

```
IF estimated_value >= 10000 THEN urgency_score += 2
ELSE IF estimated_value >= 5000 THEN urgency_score += 1
```

Rationale: High-value inventory requires more careful management.

Rule 1.4: Seasonal Context

```
IF season_match = TRUE AND weather_event = TRUE THEN urgency_score += 2
ELSE IF season_match = TRUE THEN urgency_score += 1
```

Rationale: Seasonal demand spikes are more predictable and actionable.

Rule 1.5: Final Urgency Classification

```
IF urgency_score >= 7 THEN severity = "CRITICAL"
ELSE IF urgency_score >= 4 THEN severity = "HIGH"
ELSE severity = "MEDIUM"
```

2. Action Recommendation Rules

Rule 2.1: Reorder Quantity

```
IF severity = "CRITICAL" THEN
    reorder_quantity = current_velocity * 24 * 3 // 3 days supply
ELSE IF severity = "HIGH" THEN
    reorder_quantity = current_velocity * 24 * 2 // 2 days supply
ELSE
    reorder_quantity = current_velocity * 24 * 1 // 1 day supply
```

Rationale: Higher urgency requires larger safety stock.

Rule 2.2: Reorder Priority

```
IF severity = "CRITICAL" AND hours_to_stockout <= 4 THEN
    priority = "RUSH_EXPEDITED"
ELSE IF severity = "CRITICAL" THEN
    priority = "RUSH"
ELSE IF severity = "HIGH" THEN
    priority = "STANDARD_PRIORITY"
ELSE
    priority = "STANDARD"
```

Rule 2.3: Pricing Adjustment

```
IF severity = "CRITICAL" AND velocity_ratio >= 10 THEN
    price_adjustment = 15% // surge pricing
ELSE IF severity = "HIGH" AND velocity_ratio >= 5 THEN
    price_adjustment = 10%
ELSE IF velocity_ratio >= 3 THEN
    price_adjustment = 5%
ELSE
    price_adjustment = 0%
```

Rationale: Balance demand with supply through dynamic pricing.

Rule 2.4: Stock Transfer

```
IF hours_to_stockout <= 8 AND nearby_stores_have_stock = TRUE THEN
    recommend_transfer = TRUE
    transfer_quantity = MIN(needed_quantity, available_nearby)
```

Rationale: Transfers are faster than vendor reorders for urgent needs.

Rule 2.5: Notification Escalation

```
IF severity = "CRITICAL" THEN
    notify = ["buying_team", "store_manager", "regional_manager"]
ELSE IF severity = "HIGH" THEN
    notify = ["buying_team", "store_manager"]
ELSE
    notify = ["buying_team"]
```

3. Size/Color Distribution Rules

Rule 3.1: Size Imbalance Detection

```
IF specific_size_velocity > (avg_size_velocity * 2) THEN
    recommend_size_specific_reorder = TRUE
    focus_sizes = [high_velocity_sizes]
```

Rule 3.2: Color Preference

```
IF specific_color_velocity > (avg_color_velocity * 2) THEN
    recommend_color_specific_reorder = TRUE
    focus_colors = [high_velocity_colors]
```

4. Historical Context Rules

Rule 4.1: Baseline Comparison

```
IF current_velocity > (baseline_velocity * 3) THEN
    anomaly_confirmed = TRUE
ELSE
    anomaly_confirmed = FALSE
```

Rule 4.2: Trend Analysis

```
IF velocity_increasing_for_3_hours = TRUE THEN
    trend = "ACCELERATING"
    urgency_score += 1
ELSE IF velocity_stable_for_3_hours = TRUE THEN
    trend = "STABLE"
ELSE
    trend = "DECELERATING"
    urgency_score -= 1
```

5. Constraint Rules

Rule 5.1: Vendor Capacity

```
IF vendor_max_daily_capacity < reorder_quantity THEN
    split_order = TRUE
    primary_vendor_quantity = vendor_max_daily_capacity
    secondary_vendor_quantity = reorder_quantity - vendor_max_daily_capacity
```

Rule 5.2: Budget Constraints

```
IF estimated_reorder_cost > daily_budget_remaining THEN
    require_approval = TRUE
```

```
approval_level = "REGIONAL_MANAGER"
```

Rule 5.3: Warehouse Capacity

```
IF warehouse_available_space < reorder_quantity THEN  
  adjust_reorder_quantity = warehouse_available_space  
  recommend_alternative_storage = TRUE
```

Decision Flow Example

1. Receive velocity spike alert
2. Calculate urgency_score using Rules 1.1–1.4
3. Classify severity using Rule 1.5
4. Determine reorder quantity using Rule 2.1
5. Set reorder priority using Rule 2.2
6. Calculate price adjustment using Rule 2.3
7. Check transfer options using Rule 2.4
8. Set notifications using Rule 2.5
9. Apply constraints using Rules 5.1–5.3
10. Generate recommendation

Example Calculation

Input Alert:

- SKU: WJ-2024-BLK-L
- Current velocity: 25 units/hour
- Baseline velocity: 2 units/hour
- Velocity ratio: 12.5x
- Hours to stockout: 3 hours
- Estimated value: \$6,749
- Season: Winter
- Weather event: Cold snap

Calculation:

```
urgency_score = 0
+ 3 (velocity_ratio >= 10.0)
+ 3 (hours_to_stockout <= 4)
+ 1 (estimated_value >= 5000)
+ 2 (season_match AND weather_event)
= 9 points

severity = "CRITICAL" (score >= 7)

reorder_quantity = 25 * 24 * 3 = 1,800 units
priority = "RUSH_EXPEDITED" (CRITICAL + hours <= 4)
price_adjustment = 15% (CRITICAL + ratio >= 10)
notify = ["buying_team", "store_manager", "regional_manager"]
```

Output Recommendation:

CRITICAL urgency: Rush expedited reorder of 1,800 units.
Apply 15% surge pricing. Check nearby stores for immediate transfer. Notify buying team, store manager, and regional manager.

Rule Maintenance

These rules should be reviewed and updated based on:

- Historical accuracy of predictions
- Business policy changes
- Seasonal performance data
- Vendor relationship changes
- Market conditions

Demo Limitations

For demo purposes, these rules are simplified:

- No integration with actual ERP systems
- No real-time vendor availability checks
- No actual financial transactions
- Simplified constraint handling
- Focus on explanation over execution